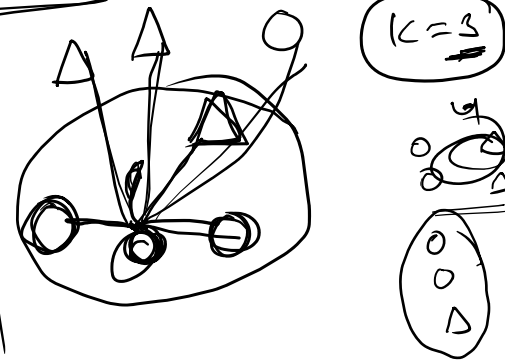
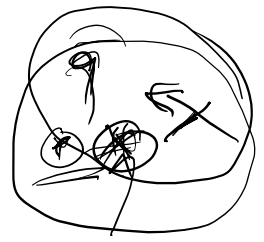


KNN $\rightarrow$ K Nearest Neighbors

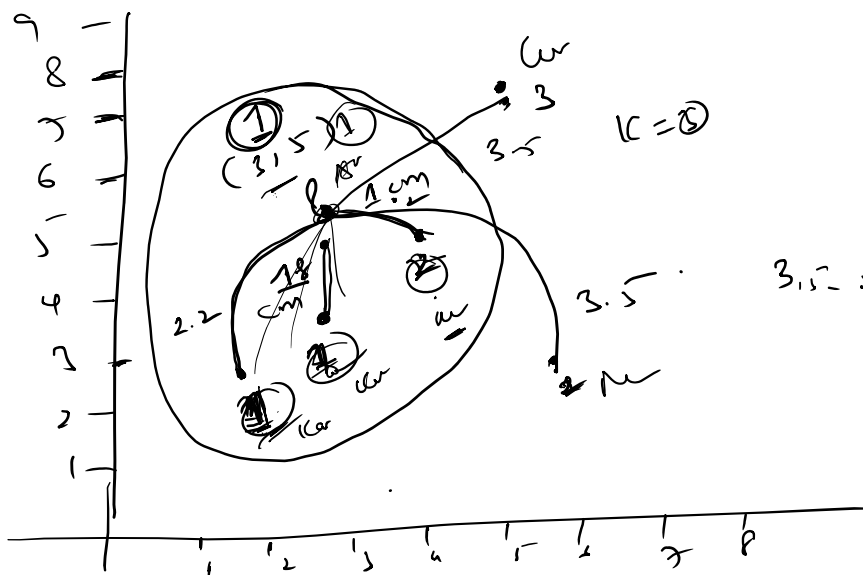
Regression

Classification



$$X = [\underline{3}, \underline{4}] [\underline{2}, \underline{3}] [\underline{4}, \underline{5}] [\underline{6}, \underline{3}] [\underline{5}, \underline{8}]$$

$$Y = [\underline{1}] [\underline{1}] [\underline{2}] [\underline{2}] [\underline{3}]$$



$$[\underline{3}, \underline{5}]$$



$$(x_1, y_1) (x_2, y_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad \text{Euclidean distance}$$

$$[\underline{3}, \underline{5}] [\underline{3}, \underline{4}] = \sqrt{(\underline{3} - \underline{3})^2 + (\underline{4} - \underline{5})^2} = \sqrt{0 + 1} = 1$$

$$(x_1, y_1) (x_2, y_2) = \sqrt{1^2 + 2^2} = \sqrt{1 + 4} = \sqrt{5} = 2.2$$

$$(3,5) \quad (2,3) = \sqrt{1^2 + 2^2} = \sqrt{1+4} = \sqrt{5} = 2.2$$

$$(3,5) \quad (4,5) = \sqrt{1^2 + 0^2} = \sqrt{1+0} = \sqrt{1} = 1$$

$$(2,5) \quad (6,3) = \sqrt{3^2 + 2^2} = \sqrt{9+4} = \sqrt{13} = 3.5$$

$$(3,5) \quad (5,8) = \sqrt{2^2 + 2^2} = \sqrt{4+4} = \sqrt{8} = 3.5$$